

RM02

## Analytical and Critical Reasoning; Syllogism

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*Syllogism and logical Venn diagram items have appeared in RAS Prelims 2021, 2023 and 2024. With cause-and-effect, fact/inference/judgement and small puzzles, this block is worth about 4-6 marks. It is pure method - there is nothing here to memorise from the world.*

Syllogism is the one reasoning topic where 'it sounds right' is always wrong.

Draw the circles, apply the rules, key the answer. This chapter gives you the four statement forms, the Venn diagram described in words, the rules for definite and possible conclusions, the either-or trick - and then the smaller cousins: logical deduction, cause and effect, fact-inference-judgement and self-contained puzzles.

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## Analytical & Critical Reasoning

### Syllogism

- Four forms: A (All), E (No), I (Some), O (Some not)
- Conversion rules
- Combination table:  $A+A=A$ ,  $A+E=E$ ,  $I+A=I$ ,  $I+E=O$
- Either-or complementary pair
- Possibility conclusions

### Logical deduction

- 'A only if B' means A implies B
- Modus ponens - P true, so Q
- Modus tollens - Q false, so not P
- Necessary vs sufficient condition

### Cause and effect

- Earlier + independent event = cause
- Both from a third event = common cause
- Unrelated spheres = independent causes

### Fact / Inference /

#### Judgement

- Fact - observed or verifiable from published data
- Inference - 'must have', 'therefore', 'is likely to'
- Judgement - 'should', 'ought', 'better than'

### Critical reasoning

- Argument = premises + conclusion
- Weaken = give an alternative explanation
- Strengthen = rule alternatives out; randomised trial is best
- Correlation is not causation

### Venn & sets

- $n(A \text{ or } B) = n(A) + n(B)$
- $n(\text{both})$
- Concentric = full inclusion
- Separate circles inside a bigger one = mutually exclusive

### Puzzles

- Ordering - build one chain with '>'
- Assignment - grid of who does what
- Truth-teller - test each case
- Input-output - what moves, and to which end

**What this chapter asks, and the one discipline that carries it**

This chapter is a family of small, self-contained items. Two or three class statements followed by two conclusions. Two events, and you must say which caused which. A sentence to be classified as fact, inference or judgement. A short paragraph whose argument you must weaken or strengthen. A Venn diagram to identify. A five-person ordering puzzle. None of them needs outside knowledge; all of them need one discipline - put the thing on paper before you decide.

- Syllogism - all/some/no statements, definite and possibility conclusions.
- Logical deduction - only if, if-then, necessary and sufficient conditions.
- Cause and effect - which of two events caused the other, or neither.
- Assertion of fact vs inference vs judgement - classify one sentence.
- Short critical reasoning passages - weaken, strengthen, find the assumption, name the flaw.
- Logical Venn diagrams and two-set counting.
- Small puzzles - ordering, assignment, truth-tellers, input-output.

**TRAP**

Never decide a syllogism by how the sentence sounds. 'All singers are dancers, some dancers are poets, therefore some singers are poets' sounds perfectly fine and is flatly wrong. The examiner writes items that sound right and are wrong. Draw the circles.

**ASKED IN RAS**

RAS Pre 2024 asked the Rajasthan-India-Asia logical Venn item (answer: three concentric circles, Rajasthan innermost). RAS Pre 2023 asked the two-set counting item - 60 students, 35 cricket, 30 football, 12 both, how many play neither (answer: 7). RAS Pre 2021 asked the English-Hindi pass percentage item (answer: 50 per cent passed both). Syllogism proper has come in all three years.

## The four statement forms - A, E, I, O

Every syllogism statement is one of exactly four types. Learn the letters, because the combination rules are written in them. A and E are universal - they speak about the whole class. I and O are particular - they speak about a part of it. A and I are affirmative; E and O are negative. Once you can label a statement in one second, the rest of the topic is mechanical.

### The four standard propositions

| Letter | Form             | Name                   |
|--------|------------------|------------------------|
| A      | All A are B      | Universal affirmative  |
| E      | No A is B        | Universal negative     |
| I      | Some A are B     | Particular affirmative |
| O      | Some A are not B | Particular negative    |

- 'Some' in logic means at least one, possibly all - it does not mean 'not all'.
- Indian competitive exams follow the existential convention: every class named is taken to be non-empty.
- That convention is why 'All A are B' is allowed to give 'Some B are A'.
- The middle term is the term common to both statements; it must not appear in the conclusion.
- A valid conclusion always joins the two END terms - the ones that appear only once.

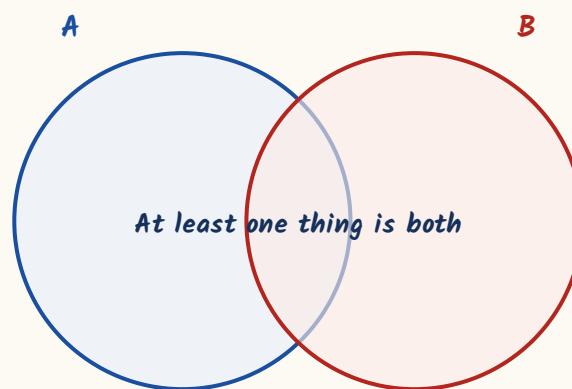
### YAAD RAKHO

Remember - A E I O comes from the Latin Affirmo and nEgO. The first two vowels of Affirmo, A and I, are the two affirmative forms; the two vowels of nEgO, E and O, are the two negatives. Universal ones come first in each pair.

## Drawing the Venn diagram in words, and the conversion rules

You do not need a neat diagram, only a correct one. Draw it as fast as you can write. All A are B is a small circle wholly inside a bigger one. No A is B is two circles that do not touch at all. Some A are B is two circles that overlap in a lens shape. Some A are not B is the same picture, but you mark a dot in the part of A that lies outside B - that dot is what the statement guarantees.

**Some A are B - the only thing guaranteed**



In logic 'some' means at least one, possibly all - it never means 'not all'.

### How to draw and read a syllogism

Label the statement A, E, I or O and find the middle term



Draw the first statement exactly - inside, apart, or overlapping



Attach the second statement to the middle term without disturbing the first



Read off what is **FORCED** in every possible drawing - that is what follows



If the second drawing could look different, the conclusion is not definite



If neither conclusion is definite, check for the either-or pair

### Valid conversions - a statement turned round

| Statement        | What it converts to                         |
|------------------|---------------------------------------------|
| Some A are B     | Some B are A (always valid)                 |
| No A is B        | No B is A (always valid)                    |
| All A are B      | Some B are A only - NEVER All B are A       |
| Some A are not B | Nothing at all - it has no valid conversion |

**TRAP**

Two conversion traps RPSC lives on. First: 'All A are B' does not become 'All B are A' - so from 'All doctors are engineers' nothing follows about a particular engineer. Second: 'Some A are not B' does not give 'Some B are not A'. Students convert it out of symmetry and lose the mark.

## The combination table - what two statements together give you

Once the two statements are labelled, their combination is fixed. Learn this table and you will key most two-statement items without drawing anything - keeping the diagram in reserve for the hard ones. Three blocking rules sit on top of it and cut the work further.

### Two statements in, one conclusion out

| Combination                     | Definite conclusion                                          |
|---------------------------------|--------------------------------------------------------------|
| All + All (A + A)               | All (A)                                                      |
| All + No (A + E)                | No (E)                                                       |
| Some + All (I + A)              | Some (I)                                                     |
| Some + No (I + E)               | Some ... are not ... (O)                                     |
| No + All, in that order (E + A) | Some ... are not ..., after conversion - watch the direction |
| Some + Some                     | No definite conclusion                                       |
| No + No                         | No definite conclusion                                       |

| Combination               | Definite conclusion                                          |
|---------------------------|--------------------------------------------------------------|
| All + Some, in that order | No definite conclusion - the middle term stays undistributed |

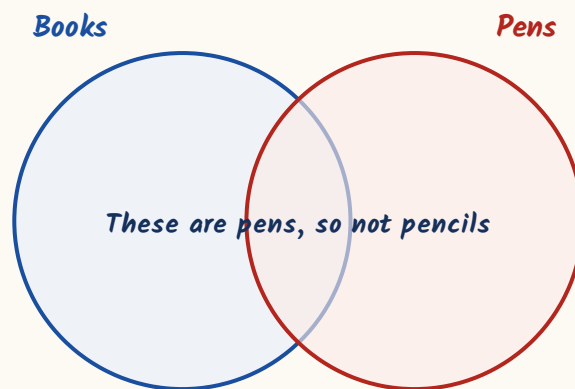
- Two particular premises (Some + Some) never give a definite conclusion.
- Two negative premises never give a definite conclusion.
- If one premise is particular, the conclusion must be particular; if one is negative, the conclusion must be negative.
- The middle term must be distributed at least once - covered by the word 'All' or 'No' somewhere.
- After finding the definite conclusion, remember its conversion also follows.

Worked example. Statements: Some books are pens. No pen is a pencil.

Conclusion I: No book is a pencil. Conclusion II: Some books are not pencils.

- Label them - 'Some books are pens' is I, 'No pen is a pencil' is E. So this is I + E, which gives O.
- Draw it: book and pen circles overlap; the pen circle sits wholly apart from the pencil circle.
- The books inside the overlap are pens, so they cannot be pencils. That gives 'some books are not pencils' - II follows.
- But the books outside the overlap are untouched by the statements - they may lie inside the pencil circle.
- So the universal 'No book is a pencil' is not forced. I does not follow.
- Answer: Only conclusion II follows.

*Some books are pens; no pen is a pencil*

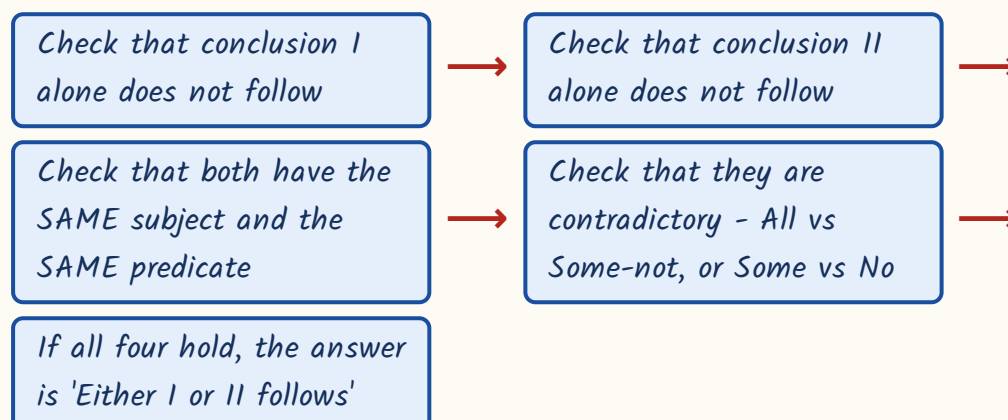


*Books outside the overlap may still be pencils, so only 'Some books are not pencils' follows.*

## ***Either-or - the complementary-pair trick***

*Sometimes neither conclusion follows on its own, yet the answer is not 'Neither'. If the two conclusions are contradictories - between them they cover every possibility, and no third case exists - then one of them must be true even though you cannot say which. That is the 'Either I or II follows' answer. It is worth learning properly, because students who do not know it key 'Neither' and lose the mark every time.*

### ***Testing for either-or***



Worked example. Statements: Some pens are books. All books are copies.

Conclusion I: All pens are copies. Conclusion II: Some pens are not copies.

- I + A gives only 'Some pens are copies' - that much is certain, because the pens that are books must be copies.
- The remaining pens are unconstrained. They may all be copies, or some may not be. So neither I nor II is forced alone.
- Now check the pair. Same subject (pens), same predicate (copies). 'All pens are copies' and 'Some pens are not copies' are contradictories.
- Between them they cover every possibility - there is no third case.
- Answer: Either I or II follows.

#### **YAAD RAKHO**

The two complementary pairs to look for, and there are only two. All A are B / Some A are not B. And Some A are B / No A is B. If the pair in front of you is not one of these two shapes, it is not an either-or item.

## **Possibility conclusions**

A possibility conclusion reads 'All A being B is a possibility' or 'Some A being B is a possibility'. The rule is the opposite of the one for definite conclusions. A definite conclusion must hold in every drawing. A possibility only needs ONE consistent drawing. So do not ask whether it is forced - ask whether it is forbidden. If you can draw even one picture where the statements are all true and the possibility holds, it follows.

- A possibility follows if at least one consistent Venn diagram allows it.
- It fails only when the statements definitely forbid it.
- A definite conclusion and its possibility-negation cannot both hold.
- So if 'No A is B' definitely follows, then 'Some A being B is a possibility' is FALSE.
- And if 'Some A are not B' definitely follows, 'All A being B is a possibility' is false.

Worked example, two items side by side. First - Statements: All apples are fruits. No fruit is a vegetable. Conclusion I: Some apples being vegetables is a possibility. Conclusion II: No apple is a vegetable. Second - Statements: All teachers are graduates. Some graduates are lawyers. Conclusion: All lawyers being teachers is a possibility.

- First item: apples sit inside fruits, and the fruit circle is wholly apart from the vegetable circle. So no apple can be a vegetable - II is definite.
- Because II is definite, the possibility in I is ruled out. Answer: Only conclusion II follows.
- Second item: is 'all lawyers are teachers' forbidden? Try to draw it - put the whole lawyer group inside the teacher circle.
- Teachers are inside graduates, so those lawyers are graduates too - 'Some graduates are lawyers' stays true, and 'All teachers are graduates' stays true.
- One consistent drawing exists, so the possibility follows.

### TRAP

Do not answer a possibility item by asking 'must this be so?' - that is the definite test and it will make you reject every possibility. Ask only 'is this forbidden?' The two questions give opposite answers, and mixing them up is the commonest error in the whole topic.

## Logical deduction - only if, if-then, necessary and sufficient

A small family of items turns on one conditional sentence. Get the direction of the arrow right and they are free marks; get it wrong and every option looks plausible. 'If P then Q' means P gives you Q, never the reverse. And 'A only if B' means B is necessary for A - so A implies B. Read that twice: the words 'only if' put the arrow the way most students do not expect.

### The four moves - two valid, two fallacies

| Given 'If P then Q', and...      | Verdict                            |
|----------------------------------|------------------------------------|
| P is true, therefore Q           | VALID - modus ponens               |
| Q is false, therefore P is false | VALID - modus tollens              |
| Q is true, therefore P           | FALLACY - affirming the consequent |
| P is false, therefore not Q      | FALLACY - denying the antecedent   |

### Necessary vs sufficient - the distinction RPSC tests

| NECESSARY condition                                                  | SUFFICIENT condition                                            |
|----------------------------------------------------------------------|-----------------------------------------------------------------|
| Without it, the thing cannot happen                                  | With it, the thing is guaranteed                                |
| Signalled by 'only if', 'only those who...'                          | Signalled by 'if', 'every ... was'                              |
| 'Only those who passed were called' - being called proves you passed | 'Everyone above 90 got a scholarship' - 85 marks proves nothing |
| Having it does not guarantee the result                              | Not having it does not rule the result out                      |
| Cleared the physical test - necessary for selection                  | Scoring above 90 - sufficient for the scholarship               |

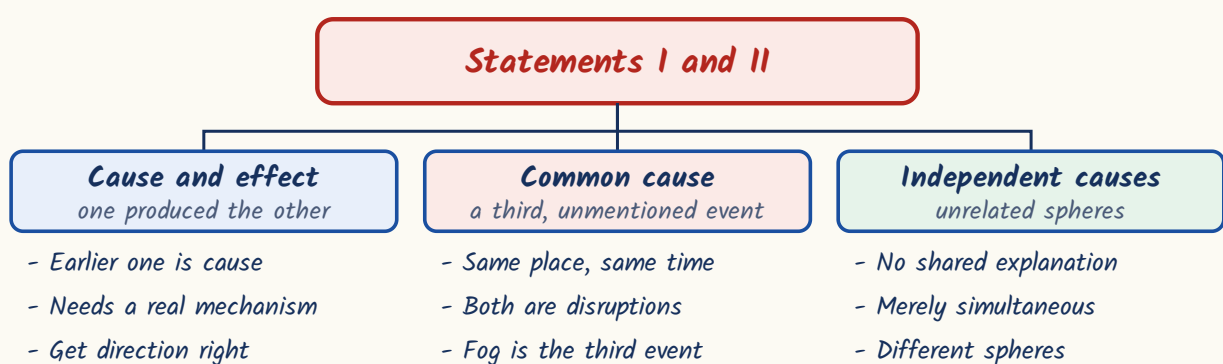
Worked example. Statement: The morning bus reaches Jaipur on time only if it leaves Jodhpur before 6 a.m. Today the morning bus reached Jaipur on time. What definitely follows?

- Name the parts.  $P$  = reaches Jaipur on time.  $Q$  = left Jodhpur before 6 a.m.
- 'P only if Q' means  $P$  implies  $Q$  - leaving early is necessary for arriving on time.
- We are told  $P$  is true today. Modus ponens gives  $Q$ : the bus left Jodhpur before 6 a.m. today.
- What does NOT follow: 'every bus that leaves before 6 a.m. reaches on time' - that is the converse, a fallacy.
- Nor does anything follow about other days - the statement was about today only.
- Answer: The bus left Jodhpur before 6 a.m. today.

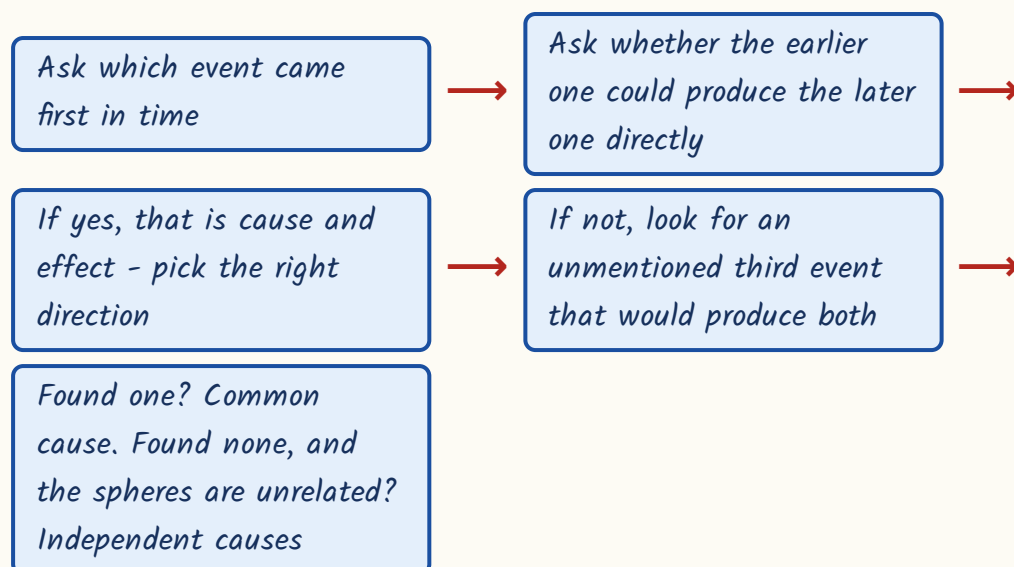
## Cause and effect, and fact vs inference vs judgement

Cause-and-effect items give you two statements and four options: I caused II, II caused I, both are effects of a common cause, or both are effects of independent causes. The deciding question is time and independence. The event that came earlier, and that could have happened on its own, is the cause. If both events plainly follow from some third event that is not mentioned, the answer is common cause. If they belong to unrelated spheres altogether, it is independent causes.

### Two events - the four possible verdicts



### Choosing among the four options



### Common cause vs independent causes

| COMMON CAUSE                                                       | INDEPENDENT CAUSES                                                         |
|--------------------------------------------------------------------|----------------------------------------------------------------------------|
| The two events are clearly linked, but not to each other           | The two events have nothing to do with each other                          |
| One unmentioned event explains both                                | No single event explains both                                              |
| Flights cancelled + schools closed on the same morning - dense fog | Gold price up in the world market + vegetable prices up in the local mandi |
| Same place, same time, same kind of disruption                     | Different spheres, merely simultaneous                                     |

### Fact, inference, judgement - decide by the signal words

| Type      | What it is                                                 | Signal words                                              |
|-----------|------------------------------------------------------------|-----------------------------------------------------------|
| Fact      | Already known, observed, or verifiable from published data | 'As per Census 2011...', a named source, a figure         |
| Inference | A conclusion drawn from the given facts                    | must have, therefore, is likely to, evidently             |
| Judgement | An opinion, approval, disapproval or recommendation        | should, must be made, ought, better than, more satisfying |

Worked example. Statement I: Many flights from the city airport were cancelled this morning. Statement II: All schools in the city were ordered to remain closed this morning.

- Test direction one: do flight cancellations close schools? No - schools do not depend on flights.
- Test direction two: does closing schools cancel flights? No - no mechanism at all.
- So it is not cause and effect either way.
- Are they unrelated? No - same city, same morning, both a disruption of normal activity. That points at a third event.
- Severe weather - dense fog or a heavy storm - would ground flights and close schools alike, and it is not mentioned.
- Answer: Both statements are effects of a common cause.

## Critical reasoning, logical Venn and the small puzzles

In a critical reasoning passage the argument is premises plus a conclusion, and the question asks you to weaken it, strengthen it, name its unstated assumption, or name its flaw. There is one master idea behind all four. Evidence supports a causal conclusion only if no other explanation fits the same evidence. So to weaken, supply an alternative explanation. To strengthen, rule alternatives out - and the strongest possible support is a controlled or randomised comparison.

- **Weaken** - show the evidence covers only part of the claim, or supply a rival cause.
- **Strengthen** - randomised assignment beats every other option, because it removes self-selection.
- **Assumption** - the unstated link the argument needs to get from its premise to its conclusion.
- **Correlation is not causation**: if a third factor accompanies both, the causal claim collapses.
- **Common flaws**: affirming the consequent, treating a necessary condition as sufficient, hasty generalisation from a small or self-selected sample, appeal to authority or tradition.

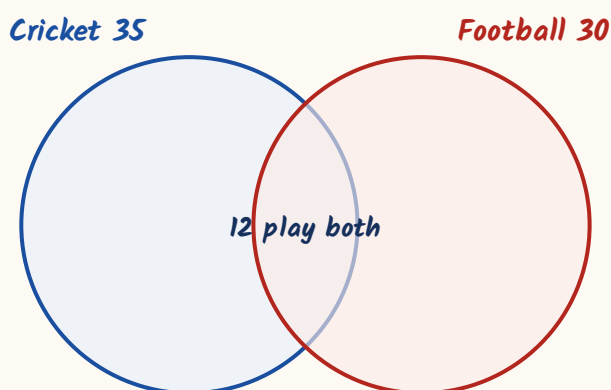
**Worked example.** A city introduced a congestion fee on cars entering its central zone. The next year the number of cars entering the zone fell by 20 per cent, and the mayor concluded that the fee had reduced car use in the city. Which fact would most weaken this?

- **Separate the parts.** Evidence: fewer cars entered the **CENTRAL ZONE**. Conclusion: car use fell in the **CITY**.
- The gap between the two is where the argument is vulnerable - a zone is not a city.
- So look for an option showing the cars went elsewhere rather than off the road.
- 'Many drivers merely shifted to roads just outside the central zone, and the total number of car trips in the city was unchanged' - that is an alternative explanation of the same evidence.
- Options about electronic collection, spending of revenue or protests say nothing about the level of car use, so they neither weaken nor strengthen.
- **Answer**: the displacement option.

**Two-set counting and logical Venn shapes**

| What is asked          | What to use                                                                       |
|------------------------|-----------------------------------------------------------------------------------|
| At least one of A or B | $n(A) + n(B) - n(\text{both})$                                                    |
| Neither A nor B        | $\text{Total} - [n(A) + n(B) - n(\text{both})]$                                   |
| Exactly one of A or B  | $n(A) + n(B) - 2 \times n(\text{both})$                                           |
| Rajasthan, India, Asia | Three concentric circles - complete inclusion, smallest innermost                 |
| Aunt, Uncle, Relative  | Two separate circles inside one bigger circle - mutually exclusive within a class |
| Doctor, Father, Man    | Overlapping circles - a person may be any combination                             |

### RAS Pre 2023 - 60 students, cricket and football



At least one =  $35 + 30 - 12 = 53$ , so neither =  $60 - 53 = 7$ .

### YAAD RAKHO

Puzzle drill, one line each. Ordering: turn every clue into a single chain with '>' before you count positions. Assignment: draw a grid of names against roles and put crosses first, ticks last. Truth-teller: test each case against every statement and keep the one that matches the given count of true statements. Input-output: identify what moves at each step and to which end - the untouched elements keep their relative order.

**ASKED IN RAS**

RAS Pre 2023 - 60 students, 35 play cricket, 30 football, 12 both. At least one =  $35 + 30 - 12 = 53$ , so neither =  $60 - 53 = 7$ . RAS Pre 2021 - 70 per cent passed English, 65 per cent Hindi, 15 per cent failed both. Passed at least one = 85, so both =  $70 + 65 - 85 = 50$  per cent. Learn the formula in this form; both papers used exactly this shape.

**REVISION RECAP**

1. Four forms - A: All A are B, E: No A is B, I: Some A are B, O: Some A are not B. Learn the letters; the rules are written in them.
2. 'Some' means at least one, possibly all; every class named is taken as non-empty.
3. Conversions: Some A are B gives Some B are A; No A is B gives No B is A; All A are B gives only Some B are A.
4. 'Some A are not B' has no valid conversion at all.
5. Combination table:  $A+A=A$ ,  $A+E=E$ ,  $I+A=I$ ,  $I+E=O$ ;  $E+A$  gives a Some-not after conversion.
6. Some + Some gives nothing; No + No gives nothing; All + Some in that order gives nothing.
7. One particular premise forces a particular conclusion; one negative premise forces a negative conclusion.
8. The middle term must be distributed at least once and must never appear in the conclusion.
9. Either-or: neither follows alone, same subject and predicate, and the two are contradictories - All/Some-not, or Some/No.
10. A possibility follows if even one consistent diagram allows it; it fails only if the statements forbid it.
11. A definite conclusion kills its own possibility-negation - if No A is B is definite, Some A being B is not possible.
12. 'A only if B' means A implies B; modus ponens and modus tollens are valid, the converse and the inverse are fallacies.
13. Necessary is not sufficient: cleared the physical test is necessary for selection, scoring above 90 is sufficient for the scholarship.
14. Cause and effect: the earlier, independently occurring event is the cause; two consequences of an unmentioned third event are effects of a common cause.

15. Events in unrelated spheres that merely coincide in time are effects of independent causes.
16. Fact is verifiable, inference carries 'must have' or 'therefore', judgement carries 'should', 'ought' or 'better than'.
17. Weaken an argument by supplying an alternative explanation; strengthen it by ruling alternatives out - a randomised comparison is the strongest support.
18.  $n(A \text{ or } B) = n(A) + n(B) - n(\text{both})$ ; neither = Total minus that; exactly one =  $n(A) + n(B) - 2 \times n(\text{both})$ .
19. Logical Venn: full inclusion is concentric circles, mutually exclusive classes within a wider class are separate circles inside one big circle.
20. In every item of this chapter, draw first and decide second - the items that sound right are the ones set to trap you.